

the clinical significance of these observations in atopic dermatitis is not known.

Via its binding to a specific cytoplasmic immunophilin (FKBP12), tacrolimus inhibits calcium-dependent signal transduction pathways in T cells, thereby preventing the

For the use only of a Registered Medical Practitioner or
a Hospital or a Laboratory

Tacroz[®]
Tacrolimus Ointment 0.1% w/w

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transcription and synthesis of IL-2, IL-3, IL-4, IL-5 and other cytokines such as GM-CSF, TNF-α and IFN-γ.

In vitro, in Langerhans cells isolated from normal human skin, tacrolimus reduced the stimulatory activity towards T cells. Tacrolimus has also been shown to inhibit the release of inflammatory mediators from skin mast cells, basophils and eosinophils.

In animals, tacrolimus ointment suppressed inflammatory reactions in experimental and spontaneous dermatitis models that resemble human atopic dermatitis. Tacrolimus ointment did not reduce skin thickness and did not cause skin atrophy in animals.

In patients with atopic dermatitis, improvement of skin lesions during treatment with tacrolimus ointment was associated with reduced Fc receptor expression on Langerhans cells and a reduction of their hyperstimulatory activity towards T cells. Tacrolimus ointment does not affect collagen synthesis in humans.

Pharmacokinetic properties

Tacrolimus concentrations in systemic circulation after topical administration are low and, when measurable, transient.

Absorption

In healthy humans, there is little or no systemic exposure to tacrolimus following single or repeated topical application of tacrolimus ointment.

Most atopic dermatitis patients (adults and children) treated with single or repeated application of tacrolimus ointment (0.03-0.1 %), and infants from age of 5 months treated with tacrolimus ointment (0.03 %) had blood concentrations < 1.0 ng/ml. When observed, blood concentrations exceeding 1.0 ng/ml were transient. Systemic exposure increases with increasing treatment areas. However, both the extent and the rate of topical absorption of tacrolimus decrease as the skin heals. In both adults and children with an average of 50% body surface area treated, systemic exposure (i.e. AUC) of tacrolimus from tacrolimus ointment is approximately 30-fold less than that seen with oral immunosuppressive doses in kidney and liver transplant patients. The lowest tacrolimus blood concentration at which systemic effects can be observed is not known.

There was no evidence of systemic accumulation of tacrolimus in patients (adults and children) treated for prolonged periods (up to one year) with tacrolimus ointment.

Distribution

As systemic exposure have been seen low with tacrolimus ointment, the high binding of tacrolimus (> 98.8%) to plasma proteins is considered not to be clinically relevant.

Following topical application of tacrolimus ointment, tacrolimus is selectively delivered to the skin with minimal diffusion into the systemic circulation.

Metabolism

Metabolism of tacrolimus by human skin was not detectable. Systemically available tacrolimus is extensively metabolised in the liver via CYP3A4.

Elimination

When administered intravenously, tacrolimus has been shown to have a low clearance rate. The average total body clearance have been seen approximately 2.25 l/h. The hepatic clearance of systemically available tacrolimus could be reduced in subjects with severe hepatic impairment, or in subjects who are co-treated with drugs that are potent inhibitors of CYP3A4.

Following repeated topical application of the ointment the average half-life of tacrolimus was estimated to be 75 hours for adults and 65 hours for children.

Paediatric population

The pharmacokinetics of tacrolimus after topical application are similar to those reported in adults, with minimal systemic exposure and no evidence of accumulation.

PHARMACEUTICAL PARTICULARS

List of excipients:

Mineral Oil, Paraffin, Propylene Carbonate, White wax, White Petrolatum

Incompatibilities

Not applicable

Shelf life

30 months

Special precautions for storage

Store below 25°C. Do not freeze. Not for children.

Nature and contents of container

A printed carton containing a leaflet and a printed lami tube containing white to slightly yellowish ointment.

Pack Size: 10g & 30g

Special precautions for disposal and other handling

No special requirements

MAL Number: MAL22036013AZ

Date of Revision:

23/09/2025

Product Registration Holder:

Glenmark Pharmaceuticals (Malaysia)
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Manufactured by:

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PHARMACEUTICALS LTD.
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® Trade Mark

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COMPOSITION

1g of Tacroz Ointment 0.1% w/w contains 1mg of tacrolimus (as tacrolimus monohydrate)

PHARMACEUTICAL FORM

Ointment. White to slightly yellowish ointment.

CLINICAL PARTICULARS

Therapeutic indications

Tacrolimus 0.03% ointment is indicated in adults, adolescents and children from the age of 2 years.

Tacrolimus 0.1 % ointment is indicated in adults and adolescents (16 years of age and above)

Flare treatment

Adults and adolescents (16 years of age and above)

Treatment of moderate to severe atopic dermatitis in adults who are not adequately responsive to or are intolerant of conventional therapies such as topical corticosteroids.

Children (2 years of age and above)

Treatment of moderate to severe atopic dermatitis in children who failed to respond adequately to conventional therapies such as topical corticosteroids.

Maintenance treatment

Treatment of moderate to severe atopic dermatitis for the prevention of flares and the prolongation of flare-free intervals in patients experiencing a high frequency of disease exacerbations (i.e. occurring 4 or more times per year) who have had an initial response to a maximum of 6 weeks treatment of twice daily tacrolimus ointment (lesions cleared, almost cleared or mildly affected).

Posology and Method of Administration

Tacrolimus treatment should be initiated by physicians with experience in the diagnosis and treatment of atopic dermatitis.

Tacrolimus is available in two strengths, Tacrolimus 0.03% and Tacrolimus 0.1% ointment.

Posology

Flare treatment

Tacrolimus can be used for short-term and intermittent long-term treatment.

Treatment should not be continuous on a long-term basis.

Tacrolimus treatment should begin at the first appearance of signs and symptoms. Each affected region of the skin should be treated with Tacrolimus until lesion are cleared, almost cleared or mildly affected. At the first signs of recurrence (flares) of the disease symptoms, treatment should be re-initiated.

Adults and adolescents (16 years of age and above)

Treatment should be started with Tacrolimus 0.1% twice a day and treatment should be continued until clearance of the lesion. If symptoms recur, twice daily treatment with Tacrolimus 0.1% should be restarted. An attempt should be made to reduce the frequency of application or to use the lower strength Tacrolimus 0.03% ointment if the clinical condition allows.

Generally, improvement is seen within one week of starting treatment. If no signs of improvement are seen after two weeks of treatment, further treatment options should be considered.

Older people

Specific studies have not been conducted in older people. However, the clinical experience available in this patient population has not shown the necessity for any dosage adjustment.

Paediatric population

Children (2 years of age and above) should use the lower strength Tacrolimus 0.03% ointment.

Treatment should be started twice a day for up to three weeks. Afterwards the frequency of application should be reduced to once a day until clearance of the lesion (see Special Warnings and Precautions for Use).

Tacrolimus ointment should not be used in children aged below 2 years until further data are available.

Maintenance treatment

Patients who are responding to up to 6 weeks treatment using tacrolimus ointment twice daily (lesions cleared, almost cleared or mildly affected) are suitable for maintenance treatment.

Adult and adolescents (16 years of age and above)

Adult patients should use Tacrolimus 0.1% ointment.

Tacrolimus ointment should be applied once a day twice weekly (e.g. Monday and Thursday) to areas commonly affected by atopic dermatitis to prevent progression to flares. Between applications there should be 2-3 days without Tacrolimus treatment.

After 12 months treatment, a review of the patient's condition should be conducted by the physician and a decision taken whether to continue maintenance treatment in the absence of safety data for maintenance treatment beyond 12 months.

If signs of a flare reoccur, twice daily treatment should be re-initiated (see Flare treatment section above).

Paediatric population

Children (2 years of age and above) should use the lower strength Tacrolimus 0.03% ointment.

Tacrolimus ointment should be applied once a day twice weekly (e.g. Monday and Thursday) to areas commonly affected by atopic dermatitis to prevent progression to flare. Between applications there should be 2-3 days without Tacrolimus treatment.

The review of the child's condition after 12 months treatment should include suspension of treatment to assess the need to continue this regimen and to evaluate the course of the disease. Tacrolimus ointment should not be used in children aged below 2 years until further data are available.

Elderly patients (65 years of age and above)

Specific studies have not been conducted in elderly patients (see Flare treatment section above).

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Method of Administration

Tacrolimus ointment should be applied as a thin layer to affected or commonly affected areas of the skin. Tacrolimus ointment may be used on any part of the body, including face, neck and flexure areas, except on mucous membranes.

Tacrolimus ointment should not be applied under occlusion because this method of administration has not been studied in patients (see Special Warnings and Precautions for Use)

Contraindications

Hypersensitivity to the active substance, macrolides in general, or to any of the excipients listed.

Special warnings and precautions for use

Exposure of the skin to sunlight should be minimised and the use of ultraviolet (UV) light from a solarium, therapy with UVB or UVA in combination with psoralens (PUVA) should be avoided during use of Tacrolimus ointment. Physicians should advise patients on appropriate sun protection methods, such as minimisation of the time in the sun, use of a sunscreen product and covering of the skin with appropriate clothing. Tacrolimus ointment should not be applied to lesions that are considered to be potentially malignant or pre-malignant.

The development of any new change different from previous eczema within a treated area should be reviewed by the physician.

The use of tacrolimus ointment is not recommended in patients with a skin barrier defect, such as Netherton's syndrome, lamellar ichthyosis, generalized erythroderma or cutaneous Graft Versus Host Disease. These skin conditions may increase systemic absorption of tacrolimus. Oral use of tacrolimus is also not recommended to treat these skin conditions. Post-marketing cases of increased tacrolimus blood level have been reported in these conditions.

Care should be exercised if applying Tacrolimus to patients with extensive skin involvement over an extended period of time, especially in children.

Patients, particularly paediatric patients should be continuously evaluated during treatment with Tacrolimus with respect to the response to treatment and the continuing need for treatment. After 12 months this evaluation should include suspension of Tacrolimus treatment in paediatric patients.

The potential for local immunosuppression (possibly resulting in infections or cutaneous malignancies) in the long term (i.e. over a period of years) is unknown.

Tacrolimus contains the active substance tacrolimus, a calcineurin inhibitor. In transplant patients, prolonged systemic exposure to intense immunosuppression following systemic administration of calcineurin inhibitors has been associated with an increased risk of developing lymphomas and skin malignancies. In patients using tacrolimus ointment, cases of malignancies, including cutaneous (i.e. cutaneous T Cell lymphomas) and other types of lymphoma, and skin cancers have been reported. Tacrolimus should not be used in patients with congenital or acquired immunodeficiencies or in patients on therapy that cause immunosuppression.

Patients with atopic dermatitis treated with Tacrolimus have not been found to have significant systemic tacrolimus levels.

The majority of these cases were related to infections (skin, respiratory tract, tooth) and resolved with appropriate antibiotic therapy. Transplant patients receiving immunosuppressive regimens (e.g. systemic tacrolimus) are at increased risk for developing lymphoma; therefore patients who receive Tacrolimus and who develop lymphadenopathy should be monitored to ensure that the lymphadenopathy resolves. Lymphadenopathy present at initiation of therapy should be investigated and kept under review. In case of persistent lymphadenopathy, the aetiology of the lymphadenopathy should be investigated. In the absence of a clear aetiology for the lymphadenopathy or in the presence of acute infectious mononucleosis, discontinuation of Tacrolimus should be considered.

Tacrolimus ointment has not been evaluated for its efficacy and safety in the treatment of clinically infected atopic dermatitis. Before commencing treatment with Tacrolimus ointment, clinical infections at treatment sites should be cleared. Patients with atopic dermatitis are predisposed to superficial skin infections. Treatment with Tacrolimus may be associated with an increased risk of folliculitis and herpes viral infections (herpes simplex dermatitis [eczema herpeticum], herpes simplex [cold sores], Kaposi's varicelliform eruption). In the presence of these infections, the balance of risks and benefits associated with Tacrolimus use should be evaluated.

Emollients should not be applied to the same area within 2 hours of applying Tacrolimus ointment. Concomitant use of other topical preparations has not been assessed. There is no experience with concomitant use of systemic steroids or immunosuppressive agents.

Care should be taken to avoid contact with eyes and mucous membranes. If accidentally applied to these areas, the ointment should be thoroughly wiped off and/or rinsed off with water.

The use of Tacrolimus ointment under occlusion has not been studied in patients. Occlusive dressings are not recommended.

As with any topical medicinal product, patients should wash their hands after application if the hands are not intended for treatment.

Tacrolimus is extensively metabolised in the liver and although blood concentrations are low following topical therapy, the ointment should be used with caution in patients with hepatic failure.

Instruct patients not to smoke or go near naked flames - risk of severe burns. Fabric (clothing, bedding, dressings etc) that has been in contact with this product burns more easily and is a serious fire hazard. Washing clothing and bedding may reduce product build-up but not totally remove it.

Immunosuppressed patients are at increased risk for opportunistic infections, including activation of latent viral infections. These include BK virus associated nephropathy which has been observed in patients receiving immunosuppressants. These infections may lead to serious, including fatal outcomes.

Interaction with other medicinal products and other forms of interaction

Formal topical drug interaction studies with tacrolimus ointment have not been conducted.

Tacrolimus is not metabolised in human skin, indicating that there is no potential for percutaneous interactions that could affect the metabolism of tacrolimus.

Systemically available tacrolimus is metabolised via the hepatic Cytochrome P4503A4 (CYP3A4). Systemic exposure from topical application of tacrolimus ointment is low (< 1.0 ng/ml) and is unlikely to be affected by concomitant use of substances known to be inhibitors of CYP3A4. However, the possibility of interactions cannot be ruled out and the concomitant systemic administration of known CYP3A4 inhibitors (e.g. erythromycin, itraconazole, ketoconazole and diltiazem) in patients with widespread and/or erythrodermic disease should be done with caution.

Paediatric population

An interaction study with protein-conjugated vaccine against Neisseria meningitidis serogroup C has been investigated in children aged 2-11 years. No effect on immediate response to vaccination, the generation of immune memory, or humoral and cell-mediated immunity has been observed.

Fertility, Pregnancy and lactation

Fertility

There are no fertility data available.

Pregnancy

There are no adequate data from the use of tacrolimus ointment in pregnant women. Studies in animals have shown reproductive toxicity following systemic administration.

The potential risk for humans is unknown.

Tacrolimus ointment should not be used during pregnancy unless clearly necessary.

Breast-feeding

Human data demonstrate that, after systemic administration, tacrolimus is excreted into breast milk. Although clinical data have shown that systemic exposure from application of tacrolimus ointment is low, breast-feeding during treatment with Tacrolimus ointment is not recommended.

Effects on ability to drive and use machines

Tacrolimus ointment has no or negligible influence on the ability to drive or use machines.

Undesirable effects

There are reports of some type of skin irritation adverse reaction at the site of application. Burning sensation and pruritus were very common, usually mild to moderate in severity and tended to resolve within one week of starting treatment. Erythema (superficial reddening of the skin) was a common skin irritation adverse reaction. Sensation of warmth, pain, paraesthesia (pins and needles) and rash at the site of application were also commonly observed. Alcohol intolerance (facial flushing or skin irritation after consumption of an alcoholic beverage) was common.

Patients may be at an increased risk of folliculitis, acne and herpes viral infections.

Adverse reactions with suspected relationship to treatment are listed below by system organ class. Frequencies are defined as very common common and uncommon within each frequency grouping. undesirable effects are presented in order of decreasing seriousness.

System Organ Class	Very Common	Common	Uncommon	Not known (cannot be estimated from the available data)
Infections and infestations		Local skin infection regardless of specific aetiology including but not limited to: Eczema herpeticum, Folliculitis, Herpes simplex, Herpes virus infection, Kaposi's varicelliform eruption*		Ophthalmic Herpes Infection*
Metabolism and nutrition disorders		Alcohol intolerance (facial flushing or skin irritation after consumption of an alcoholic beverage)		
Nervous system disorders		Paraesthesias and dysaesthesias (hyperaesthesia, burning sensation)		
Skin and subcutaneous tissue disorders		Pruritus	Acne*	Rosacea* Lentigo*
General disorders and administration site conditions	Application site burning, Application site pruritus	Application site warmth, Application site erythema, Application site pain, Application site irritation, Application site paraesthesia, Application site rash		Application site oedema*
Investigations				Drug level increased*

* The adverse reaction has been reported during post-marketing experience

Post-marketing

Cases of malignancies, including cutaneous (i.e. cutaneous T Cell lymphomas) and other types of lymphoma, and skin cancers, have been reported in patients using tacrolimus ointment.

Reporting of suspected adverse reactions

Reporting suspected adverse reactions after authorisation of the medicinal product is important. It allows continued monitoring of the benefit/risk balance of the medicinal product. Healthcare professionals are asked to report any suspected adverse reactions.

Overdose

Overdosage following topical administration is unlikely.

If ingested, general supportive measures may be appropriate. These may include monitoring of vital signs and observation of clinical status. Due to the nature of the ointment vehicle, induction of vomiting or gastric lavage is not recommended.

PHARMACOLOGICAL PROPERTIES

Pharmacodynamic properties

Pharmacotherapeutic group: Other dermatologicals, ATC code: D11AH01

Mechanism of action and pharmacodynamic effects

The mechanism of action of tacrolimus in atopic dermatitis is not fully understood. While the following have been observed,

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